RESEARCH ARTICLE | DECEMBER 17 2020

Eigenmode analysis of the sheared-flow Z-pinch FREE

J. R. Angus ; J. J. Van De Wetering ; M. Dorf ; V. I. Geyko

Phys. Plasmas 27, 122108 (2020)
https://doi.org/10.1063/5.0029716

Physics of Plasmas ARTICLE scitation.org/journal/php

Eigenmode analysis of the sheared-flow Z-pinch

Cite as: Phys. Plasmas **27**, 122108 (2020); doi: 10.1063/5.0029716
Submitted: 15 September 2020 · Accepted: 17 November 2020 ·
Published Online: 17 December 2020

J. R. Angus,[1,a)] J. J. Van De Wetering,[2] M. Dorf,[1] and V. I. Geyko[1]

AFFILIATIONS

[1]Lawrence Livermore National Laboratory, 7000 East Avenue, L-227 Livermore, California 94550, USA
[2]Department of Physics, Princeton University, Princeton, New Jersey 08544, USA

[a)]Author to whom correspondence should be addressed: angus1@llnl.gov

ABSTRACT

Experiments have demonstrated that a Z-pinch can persist for thousands of times longer than the growth time of global magnetohydrodynamic (MHD) instabilities such as the $m = 0$ sausage and $m = 1$ kink modes. These modes have growth times on the order of $t_a = a/v_i$, where v_i is the ion thermal speed and a is the pinch radius. Axial flows with $du_z/dr \lesssim v_i/a$ have been measured during the stable period, and the commonly accepted theory is that this amount of shear is sufficient to stabilize these modes as predicted by numerical studies using the ideal MHD equations. However, these studies only consider specific equilibrium profiles that typically have a modest magnitude for the logarithmic pressure gradient, $q_P \equiv d\ln P/d\ln r$, and may not represent experimental conditions. Linear stability of the sheared-flow Z-pinch is studied here via a direct eigen-decomposition of the matrix operator obtained from the linear ideal MHD equations. Several equilibrium profiles with a large variation of q_P are examined. Considering a practical range of k, $1/3 \lesssim ka \lesssim 10$, it is shown that the shear required to *stabilize* $m = 0$ modes can be expressed as $du_z/dr \geq C\gamma_0/(ka)^\alpha$. Here, $\gamma_0 = \gamma_0(ka)$ is the profile-specific growth rate in the absence of shear, which scales approximately with $|q_P|$. Both C and α are profile-specific constants, but C is order unity and $\alpha \approx 1$. It is further demonstrated that even a large value of shear, $du_z/dr = 3v_i/a$, is not sufficient to provide linear stabilization of the $m = 1$ kink mode for all profiles considered. This result is in contrast to the currently accepted theory predicting stabilization at much lower shear, $du_z/dr = 0.1v_i/a$, and suggests that the experimentally observed stability cannot be explained within the linear ideal-MHD model.

I. INTRODUCTION

The Z-pinch configuration is an attractive magnetic-confinement concept for a compact thermonuclear fusion device because of its simple geometry and efficient utilization of magnetic field (plasma beta order unity), and it does not require any external magnetic field coils.[1] The classical Z-pinch configuration consists of a cylindrical plasma column carrying an axial current. The axial current both produces and interacts with an azimuthal magnetic field to provide a radially inward force that balances the radially outward thermal pressure gradient of the plasma. In equilibrium, the plasma pressure scales with the square of the plasma current. Assuming the system remains stable, thermonuclear fusion conditions can be reached by scaling up the current.[2]

The quasistatic Z-pinch was one of the earliest concepts considered for a thermonuclear reactor. However, the original experiments were plagued by global, fast-growing magnetohydrodynamic (MHD) instabilities that limited the lifetime of the plasma,[3,4] namely, the $m = 0$ sausage and $m = 1$ kink modes. These modes have growth times on the order of $t_a = a/v_i$, where v_i is the ion thermal speed and a is the pinch radius. Despite this, there has been a lot of experimental success in recent years illustrating that a Z-pinch plasma can be stable for thousands of times longer than t_a. This includes experiments on the ZaP (Z-Pinch) device,[5,6] which reported pinch currents of around 50 kA, and experiments on the scaled-up FuZE[7] (fusion Z-pinch experiment) device that has operated at currents exceeding 200 kA.

The current hypothesis to explain the long-time stabilization of the ZaP and FuZE plasmas is that the MHD modes that have limited past experiments are mitigated by the presence of a sheared axial flow of the plasma column.[7] Sheared axial flows u_z with $du_z/dr \lesssim v_i/a$ have been measured on ZaP experiments by looking at Doppler shifted impurity lines.[6,8] Based on computational studies, it is believed that this amount of shear can stabilize the linear modes in a Z-pinch. In particular, the computational work by Shumlak and Hartman[9] is cited for shear required to stabilize $m = 1$, and the computational work by Paraschiv *et al.*[10] is typically cited for the shear required to stabilize $m = 0$ modes. However, these numerical studies only consider specific equilibrium profiles that may not represent experimental conditions. Moreover, a puzzling feature of these models is that the trends of the required shear needed for stabilization with axial wavenumber k differ between $m = 0$ and $m = 1$.

A linear stability study of the $m = 1$ kink mode using the ideal MHD equations is done in Ref. 9. The marginally $m = 0$ stable Kadomtsev profile is considered along with an axial velocity that varies linearly with the radial coordinate r. It is determined that $m = 1$ kink modes are stable when $du_z/dr \gtrsim 0.1kV_A^0$, where k is the axial wavenumber and $V_A^0 \approx v_i$ is a characteristic Alfvén speed of the plasma. However, this result is in disagreement with essentially identical computational work done around the same time by Arber and Howell[11] that showed even a strongly sheared flow with $du_z/dr > v_i/a$ has only a modest effect on the $m = 1$ kink mode for $ka = 10/3$. The strong disparity between these two results casts doubt on the role that shear plays on the linear stability of $m = 1$ kink modes, as is discussed in two independent review articles by Terry[12] and Haines.[13]

The effects of a sheared axial flow on the stability of $m = 0$ sausage modes are studied using fully nonlinear simulations of the ideal MHD equations by Paraschiv *et al.* in Ref. 10 considering Bennett equilibrium profiles. The main result of the work in Ref. 10 is that flows such that $du_z/dr \lesssim v_i/a$ are sufficient to stabilize all $m = 0$ modes with $1/3 \lesssim ka$. It is also shown in Ref. 10 that higher-ka modes are more susceptible to shear, which is the opposite trend to the scaling with k for shear stabilization of $m = 1$ modes reported in Ref. 9.

A linear study of sheared-flow stabilization of $m = 0$ sausage modes has recently been done using fully kinetic particle-in-cell (PIC) simulations.[14] The plasma parameters for temperature and density used in this work are taken to align with those in the FuZE device. The result of this work is in agreement with the results in Ref. 10—stabilization of the $m = 0$ sausage mode can be obtained with $du_z/dr \lesssim v_i/a$. However, results are only reported for a single axial wavelength in Ref. 14 ($ka = 5$), and they also only considered Bennett equilibrium profiles.

An eigenmode analysis of the sheared-flow Z-pinch configuration is performed in the present paper via a direct eigendecomposition of the matrix operator obtained from the linear ideal MHD equations. Both Kadomtsev and Bennett profiles are considered, as well as a new class of profiles that can be fit to experimentally measured quantities from the FuZE device.[7] This latter profile can be distinguished from the other two as it has a relatively larger logarithmic pressure gradient magnitude, $|q_P \equiv d\ln P/d\ln r|$. The effects of a linearly sheared axial velocity on linear stability for each set of equilibrium profiles are quantified as a function of normalized axial wavenumber ka. It is shown that the amount of shear required to *stabilize* $m = 0$ modes scales directly with the shear-free growth rate and has an inverse scaling with ka. The shear-free growth rate scales approximately with the characteristic value of $|q_P|$. The effect of shear on $m = 1$ modes is less sensitive to the equilibrium profiles, as is the shear-free growth rate. However, even for $du_z/dr = 3v_i/a$, the growth rates for $m = 1$ and $ka \approx 1$ are *not* strongly suppressed. Furthermore, similar to $m = 0$ modes, higher ka modes are also more strongly affected by shear for $m = 1$.

The results here for $m = 1$ are in contrast to the currently accepted theory for $m = 1$ from Ref. 9 discussed above that says $du_z/dr \gtrsim 0.1kv_i$ is sufficient for stabilization of these modes. On the other hand, our results are in agreement with those from Ref. 11, and the main conclusion of this work is also the same as that from Ref. 11—if flows with $du_z/dr \lesssim v_i/a$ are responsible for the long stabilization times of Z-pinch experiments, then it should be a nonlinear process or related to non-ideal/kinetic effects. The results presented here are shown to be in agreement with analytic models obtained from the governing equations in certain limiting cases. Additionally, heuristic arguments are presented to explain the observed relationship between shear, equilibrium profiles, and axial wavenumber.

II. IDEAL MHD EQUATIONS AND EQUILBRIUM PROFILES

The ideal MHD equations in non-conservative form are

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \tag{1}$$

$$\rho \frac{d\mathbf{u}}{dt} = \mathbf{J} \times \mathbf{B} - \nabla P, \tag{2}$$

$$\frac{dS}{dt} = 0, \tag{3}$$

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0. \tag{4}$$

In the above equations, ρ is the mass density, $\mathbf{u}$ is the mean velocity, $\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$ is the current density, $\mathbf{B}$ is the magnetic field, P is the plasma pressure, $S = P/\rho^\Gamma$ is the specific entropy with Γ the adiabatic coefficient, $\mathbf{E} = -\mathbf{u} \times \mathbf{B}$ is the electric field, and $d/dt = \partial/\partial t + \mathbf{u} \cdot \nabla$.

The equilibrium condition for an infinitely long Z-pinch is radial force balance

$$\frac{dP}{dr} + \frac{B_\theta}{r\mu_0} \frac{d}{dr}(rB_\theta) = 0, \tag{5}$$

where B_θ is the azimuthal magnetic field associated with an axial current density J_z. Any arbitrary axial flow of the form $u_z = u_z(r)$ does not change the equilibrium condition given in Eq. (5).

Using variational techniques and considering perturbations that depend on an azimuthal mode number $m = 0, 1, 2, \ldots$, the following stability conditions for a Z-pinch with no equilibrium flow can be derived:[15]

$$-\frac{r}{P}\frac{dP}{dr} \le \frac{4\Gamma}{2 + \Gamma\beta}, \quad \text{for } m = 0, \tag{6}$$

$$-\frac{r}{P}\frac{dP}{dr} \le \frac{m^2}{\beta}, \quad \text{for } m > 0, \tag{7}$$

where $\beta = 2\mu_0 P/B_\theta^2$ is the equilibrium plasma beta. The above conditions are known as the Kadomtsev stability conditions for sausage ($m = 0$) and kink ($m > 0$) modes.

Two of the most well-known solutions to Eq. (5) are the Kadomtsev and Bennett profiles. The Kadomtsev profiles, which are the $m = 0$ marginally stable profiles, are a function of Γ and can be expressed parametrically[13] using Eqs. (5) and (6) with the equal sign. The most widely used set of profiles is the Bennett profiles. These profiles can be explicitly expressed as

$$P = \frac{2\bar{P}a^4}{(a^2 + r^2)^2}, \quad B_\theta = \frac{2\bar{B}ar}{a^2 + r^2}, \tag{8}$$

where a is a characteristic pinch radius, $\bar{P}$ is the characteristic ion pressure, and $\bar{B}$ is the characteristic magnetic field. An interesting feature of the Bennett profiles is that the $E \times B$ drift speed of the electrons and ions is radially uniform. Also, these profiles are equal to the Kadomtsev profiles for $\Gamma = 2$. Bennett profiles are unstable at all r in accordance with Eq. (6) for $\Gamma < 2$ and vice versa for $\Gamma > 2$ (see Ref. 16).

Measurements of the plasma density and temperature profiles from the FuZE device are presented in Ref. 7. The corresponding pressure profile can be fit to an expression of the form

$$P = P_{00} + \sum_{n=1}^{3} P_n \exp\left(-a_n r^2 - b_n r^4\right). \tag{9}$$

The magnetic field solution from Eq. (5) using P from Eq. (9) can be expressed analytically in terms of *Error* functions. This, along with the specific values of the constants in Eq. (9), is given in Appendix A. These profiles are referred to in this work as FuZE-like profiles.

The pressure and magnetic field profiles corresponding to the three sets of equilibrium discussed above are illustrated in Fig. 1. The Kadomtsev profile is obtained using $\Gamma = 5/3$. The Kadomtsev and Bennett profiles are referred to as *diffuse* profiles because their pressure gradients are gradual and B_θ slowly approaches the vacuum scaling with increasing r. On the other hand, B_θ for the FuZE-like profiles quickly approaches the vacuum scaling as the pressure is more localized within the characteristic pinch radius a.

The characteristic pinch scale a is defined to be the radial location where B_θ is maximum. The magnetic field is normalized such that $rB_\theta|_{r\to\infty} = \mu_0 I/2\pi = 2a\bar{B}$. The corresponding pressure scale is $\bar{P} = \bar{B}^2/\mu_0$. The temperature is assumed radially uniform and the same constant value for each set of profiles. For uniform temperature and infinite radial domain, the plasma temperature can be obtained by multiplying Eq. (5) by r^2 and integrating to give the well-known Bennett relation

$$T = \frac{\mu_0 I^2}{16\pi N}, \tag{10}$$

where T is the plasma temperature, $I = 2\pi\int_0^\infty J_z r dr$ is the plasma current, and $N = 2\pi\int_0^\infty n r dr$ is the plasma line density. Only the radially decaying part of the density enters N in Eq. (10) for profiles that contain a uniform background density. Thus, for equal temperatures and total current, each set of profiles has the same line density via Eq. (10).

The ion thermal speed is $v_i = \sqrt{T/M_i}$ with M_i the ion mass. The characteristic ion-transit time scale is $t_a = a/v_i$. The velocity profiles used in this work are characterized by the flow speed relative to v_i at $r = a$ and is denoted by κ. Unless stated otherwise, the equilibrium velocity profile considered in this work is linear with r and given by

$$u_z = \kappa \frac{r}{a} v_i. \tag{11}$$

A linearly varying velocity profile is typically used when studying stabilizing effects of sheared-flow in Z-pinch plasmas[9,10,14,17,18] as well as Rayleigh–Taylor[19,20] modes in electrostatic systems and fluids because it avoids introducing Kelvin–Helmholtz modes that require the second derivative of u_z to be nonzero.[21]

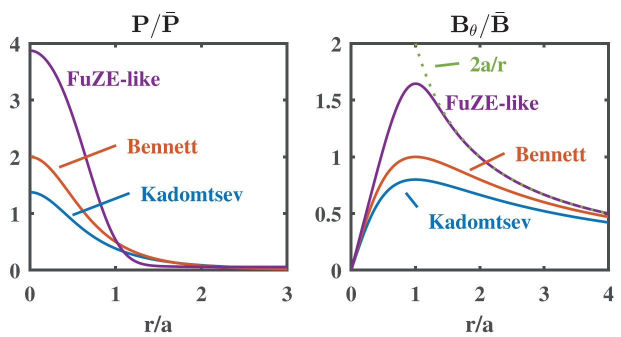

FIG. 1. Equilibrium profiles for pressure (left) and magnetic field (right) for Kadomtsev profiles with $\Gamma = 5/3$ (blue), Bennett profiles (red), and FuZE-like profiles (purple).

III. LINEAR EQUATIONS AND NUMERICAL TOOL

We are interested in normal-mode solutions of the linear MHD equations for Z-pinch equilibrium. Each variable is expressed as $f(r,\theta,z,t) = f_0(r) + f_1(r,\theta,z,t)$ where it is assumed $|f_1| \ll |f_0|$ and $f_1 = \tilde{f}(r)\exp(-i\omega t + ikz + im\theta)$. The linear form of Eqs. (1)–(4) can be written as

$$-i\omega'\tilde{P} = -\frac{dP_0}{dr}\tilde{u}_r - \Gamma P_0 \nabla\cdot\tilde{\mathbf{u}}, \tag{12}$$

$$-i\omega'\tilde{u}_r = -\frac{J_{z0}}{\rho_0}\tilde{B}_\theta - \tilde{J}_z\frac{B_{\theta 0}}{\rho_0} - \frac{1}{\rho_0}\frac{d\tilde{P}}{dr}, \tag{13}$$

$$-i\omega'\tilde{u}_\theta = \frac{J_{z0}}{\rho_0}\tilde{B}_r - \frac{im}{r\rho_0}\tilde{P}, \tag{14}$$

$$-i\omega'\tilde{u}_z = -\frac{du_{z0}}{dr}\tilde{u}_r + \tilde{J}_r\frac{B_{\theta 0}}{\rho_0} - \frac{ik}{\rho_0}\tilde{P}, \tag{15}$$

$$-i\omega'\tilde{B}_r = \frac{im}{r}B_{\theta 0}\tilde{u}_r, \tag{16}$$

$$-i\omega'\tilde{B}_\theta = -ikB_{\theta 0}\tilde{u}_z - \frac{d}{dr}\left(B_{\theta 0}\tilde{u}_r\right), \tag{17}$$

$$-i\omega'\tilde{B}_z = \frac{du_{z0}}{dr}\tilde{B}_r + \frac{im}{r}B_{\theta 0}\tilde{u}_z, \tag{18}$$

where $\omega' = \omega - ku_{z0}$ is the local Doppler shifted frequency. Equation (12) is obtained by combining the continuity equation from Eq. (1) with the entropy density equation from Eq. (3). Equation (18) is found from the z-component of Eq. (4) and making use of $\nabla\cdot\mathbf{B} = 0$. The divergence of the linear velocity is

$$\nabla\cdot\tilde{\mathbf{u}} = \frac{1}{r}\frac{d}{dr}(r\tilde{u}_r) + \frac{im}{r}\tilde{u}_\theta + ik\tilde{u}_z. \tag{19}$$

The r and z components of the linear current density are

$$\mu_0\tilde{J}_r = \frac{im}{r}\tilde{B}_z - ik\tilde{B}_\theta, \tag{20}$$

$$\mu_0\tilde{J}_z = \frac{1}{r}\frac{d}{dr}\left(r\tilde{B}_\theta\right) - \frac{im}{r}\tilde{B}_r. \tag{21}$$

The above system of linear equations, Eqs. (12)–(18), can be written in the following eigenvalue problem format:

$$\mathcal{A}\mathbf{x} = -i\omega\mathbf{x} \tag{22}$$

with eigenvector $\mathbf{x} = [\tilde{P}, \tilde{\mathbf{u}}, \tilde{\mathbf{B}}]^T$ and eigenvalues $-i\omega$. The normal modes of the system are obtained numerically in this work by a direct eigen-decomposition of the operator $\mathcal{A}$ in Eq. (22) where the elements of $\mathcal{A}$ are expressed on a discrete grid. For reasons that will be discussed below concerning boundary conditions, the $\tilde{u}_r$ and $\tilde{B}_\theta$ elements of the eigenvector $\mathbf{x}$ are replaced with $r\tilde{u}_r$ and $r\tilde{B}_\theta$. The matrix $\mathcal{A}$ is correspondingly adjusted to reflect this change. The decomposition has been performed using the *eig* operator in MATLAB as well as the *eig* operator from Python's *SciPy* linear algebra library. Both codes produce indistinguishable results.

The radial domain goes from $r=0$ to $r=R$ and is uniformly spaced with $\Delta r = R/n_r$. Convergence for all results reported in this work is achieved using $n_r = 1024$. The values are defined at cell center such that the first element on the grid exists at $\Delta r/2$. Note that the square matrix $\mathcal{A}$ has length $7n_r$ for $m > 0$. For $m=0$, the length reduces to $4n_r$ as $\tilde{B}_r = \tilde{B}_z = \tilde{u}_\theta = 0$. The derivative operators are computed using the fourth-order central scheme (*C*4). The stencils used near the boundaries are obtained using biased fourth order expressions that take into account the boundary conditions at $r=0$ and $r=R$.

A. Boundary conditions

The linear system of equations contains first order radial derivatives of $r\tilde{u}_r$, $r\tilde{B}_\theta$, and $\tilde{P}$. The $r=R$ boundary is considered as a thermally insulating, perfectly conducting wall such that $r\tilde{u}_r = d\tilde{P}/dr = d(r\tilde{B}_\theta)/dr = 0$ at $r=R$. The boundary conditions for these variables at $r=0$ can be determined by writing the coefficients in Eqs. (12)–(18) using a power series expansion for small r. Boundary condition at $r=0$ for the r-weighted Lagrangian radial displacement element, $Y \equiv r\xi_r^L = r\tilde{u}_r/(-i\omega')$, and the total pressure perturbation, $Z \equiv \tilde{P} + \tilde{B}_\theta B_{\theta 0}/\mu_0$, are derived in Ref. 22. This derivation is reviewed here because the boundary conditions at $r=0$ have been a topic of controversy in the past for a certain class of equilibrium velocity profiles.[17]

After careful algebraic manipulation, Eqs. (12)–(18) can be reduced to two coupled first order ordinary differential equations for Y and Z.[11,22] These equations can be written as

$$ASr\frac{dY}{dr} = rC_1 Y - r^2 C_2 Z, \tag{23}$$

$$ASr\frac{dZ}{dr} = C_3 Y - rC_1 Z. \tag{24}$$

The definitions of the coefficients A, S, rC_1, C_2, and C_3 are given in Appendix B. Previous authors have studied Z-pinch stability via numerical solution of these two equations using an iterative shooting method.[11] The normal modes are obtained in a different way in this work, but nevertheless the reduced system [Eqs. (23) and (24)] is useful for understanding the appropriate behavior of the solution near $r=0$ as well as for further analysis of the results to be given later in the paper.

Assuming solutions at small r of the form

$$Y = r^s\left(Y_0 + Y_1 r + Y_2 r^2 + \cdots\right), \tag{25}$$

and similarly for $Z \sim r^t$, it can be shown (based on the regularization constraint) using the power series expansion for the coefficients that $s=2$ and $t=0$ for $m=0$ and $s=t=m$ for $m>0$ (see Ref. 22). To leading order, the solution at small r thus behaves as

$$Y \sim r^2,\ Z \sim r^0, \quad \text{for } m = 0,$$
$$Y \sim Z \sim r^m, \quad \text{for } m > 0.$$

A direct inspection of Eq. (13) for small r shows that $r\tilde{u}_r$ and $r\tilde{B}_\theta$ behave the same for all m. The boundary conditions used at $r=0$ in our algorithm are that $r\tilde{u}_r$, $r\tilde{B}_\theta$, and $\tilde{P}$ have zero gradient for even m and zero value for odd m.

A subtle, but important point to make about the above expansion, is that as long as the equilibrium velocity profile is even about $r=0$, the power-law expansions for each of the coefficients (AS, rC_1, r^2C_2, and C_3) in Eqs. (23) and (24) are also even and so the odd terms in the solution, such as Y_1 and Z_1, are identically zero. For $m=1$ and even u_{z0}, $\tilde{u}_r$ can be written as

$$\tilde{u}_r = \tilde{u}_{r0} + \tilde{u}_{r2}r^2 + \cdots \tag{26}$$

in which case a zero gradient boundary condition for $\tilde{u}_r$ suffices.[9] However, if the equilibrium velocity profile is odd about the axis, then the full power-law series with both even and odd terms should be retained. The solution for $m=1$ and u_{z0} odd goes like

$$\tilde{u}_r = \tilde{u}_{r0} + \tilde{u}_{r1}r^1 + \tilde{u}_{r2}r^2 + \cdots. \tag{27}$$

The boundary condition for this situation is not well-posed for $\tilde{u}_r$ as neither its value nor its first derivative is known *a priori*.[17] The boundary condition is well posed for $r\tilde{u}_r$, which is used in this work and analogous to that used in other works.[11,22] The same arguments here could potentially be used to say that $d\tilde{P}/dr$ is finite at $r=0$ for $m=0$. However, it can be shown that the first order in r term for $\tilde{P}$ is zero for $m=0$ independent of whether u_{z0} is odd or even.

IV. RESULTS FOR $m=0$ SAUSAGE MODES

Results for $m=0$ modes are presented in this section using $R=4a$ and $\Gamma = 5/3$. In fact, *all* simulation results presented in this paper are obtained using $\Gamma = 5/3$. All resolvable eigenmodes on the grid are obtained and can be identified using an effective radial mode number n. Radial profiles of the pressure perturbation corresponding to the four fastest growing modes using Bennett profiles with $ka = 10/3$ are shown in Fig. 2. Each higher value of n corresponds to an additional oscillation of the mode structure. For fixed ka, the lowest radial mode number ($n=0$) is always the fastest growing mode

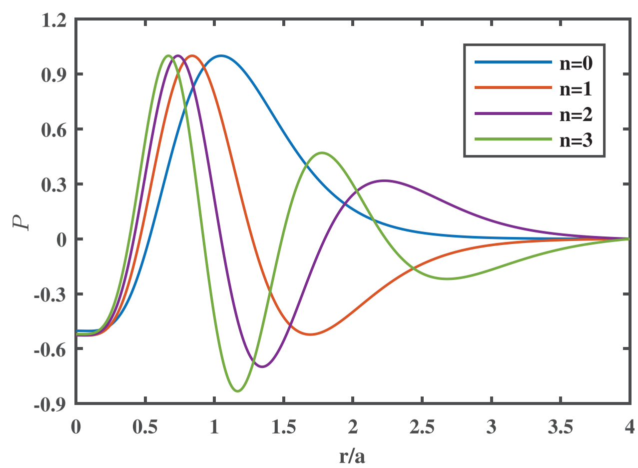

FIG. 2. Pressure perturbations corresponding to the first four radial mode numbers n for $m=0$ sausage mode with $ka = 10/3$ using Bennett profiles and $u_{z0} = 0$.

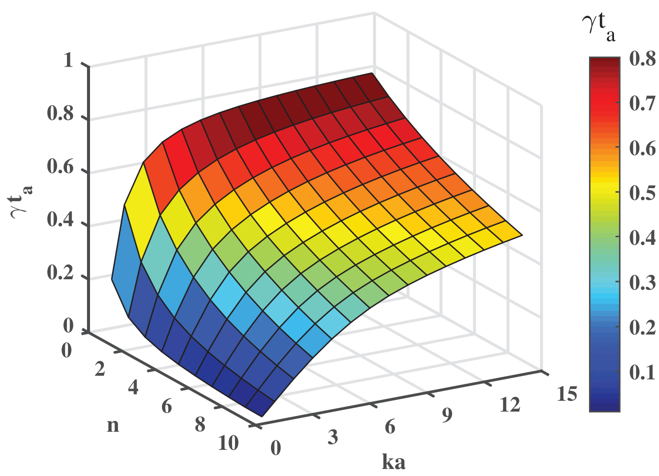

FIG. 3. Shear-free growth rates of $m=0$ sausage modes in axial wavenumber (ka) and radial mode number (n) phase space using Bennett profiles.

and the growth rates decrease monotonically with increasing n. The full growth rate spectrum of shear-free sausage modes obtained using Bennett profiles is presented in Fig. 3. The values here for $m=0$ and $n=0$ are in agreement with those obtained from nonlinear MHD simulations in Refs. 10 and 16. Only results for the fastest growing modes ($n=0$) will be presented for the remainder of this work unless stated otherwise.

The effect of a linearly sheared axial velocity on the growth rate spectrum of $m=0$ sausage modes using Bennett profiles is illustrated in Fig. 4. The non-dimensional shear strength κ is varied from $\kappa=0$ to $\kappa=0.4$. The growth rate saturates to a constant for $ka \gg 1$ for $\kappa=0$. Higher-ka modes are more strongly affected by the shear, shifting the maximum unstable mode to lower values of ka as κ is increased. The maximum growth rate is $\gamma \approx 0.3/t_a$ at $ka \approx 1$ for $\kappa=0.4$.

The effect of axial velocity shear on the growth rate spectrum of $m=0$ sausage modes using FuZE-like profiles is illustrated in Fig. 5.

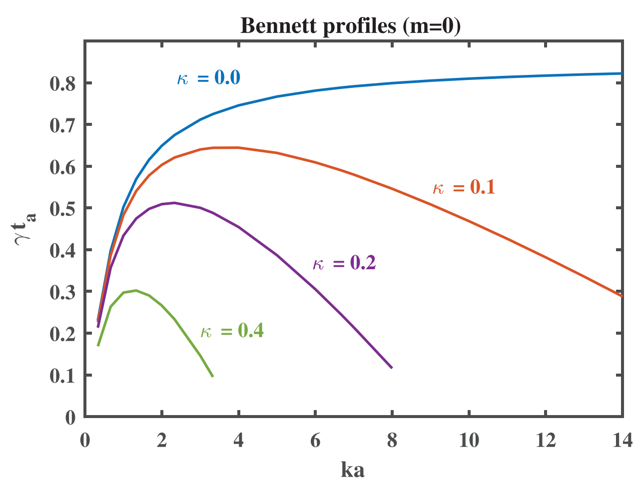

FIG. 4. Growth rate spectrum of fastest growing $m=0$ modes for different values of shear using Bennett profiles with $u_{z0}/v_i = \kappa r/a$.

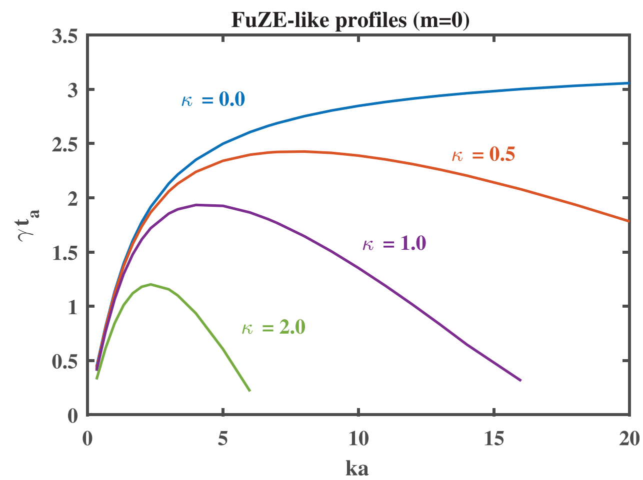

FIG. 5. Growth rate spectrum of fastest growing $m=0$ modes for different values of shear using FuZE-like profiles with $u_{z0}/v_i = \kappa r/a$.

The same suppression effect of the growth rate with increasing shear observed for Bennett profiles is also observed for these profiles. However, a shear coefficient about five times larger than that used for Bennett profiles is required to have the same relative level of suppression. The maximum growth rate is $\gamma \approx 1.1/t_a$ at $ka \approx 2$ for $\kappa = 2.0$.

A sheared axial flow causes the eigenmodes for $m=0$ to stretch axially and compress radially. This is illustrated in Fig. 6 by showing contours of the perturbed pressure, P_1, for both Bennett and FuZE-like profiles for several different values of κ. Similar to the effects of shear on the growth rate spectrum for these two sets of profiles, a 5× increase in the shear is needed to have a similar relative distortion of the eigenmode for FuZE-like profiles in comparison to Bennett profiles.

A quantity of interest is how much shear is required to suppress the growth rate to zero. It has been conjectured that shear never

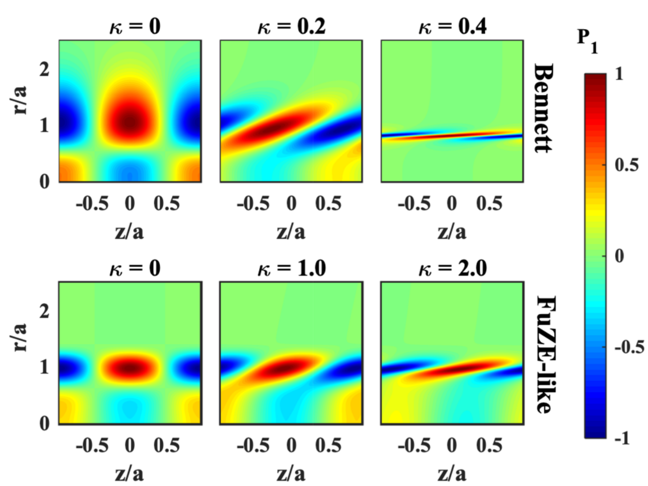

FIG. 6. Effect of non-dimensional shear strength κ on pressure perturbation for $m=0$ sausage modes using Bennett (top) and FuZE-like (bottom) profiles with $ka = 10/3$ and $u_{z0}/v_i = \kappa r/a$.

eliminates the instability completely for Rayleigh–Taylor modes in fluids.[19] It is likely that the same is true for the instabilities studied here. In either case, the value of κ such that the growth rate is zero cannot be determined from our numerical algorithm for the following reason. As the shear is increased for a fixed ka, the eigenmode gets stretched axially and compressed radially (see Fig. 6). At some point the mode becomes so compressed radially that it cannot be resolved on the grid. For $m=0$ modes with the grid spacing used in this work, a converged result for the eigenmode is not obtained for $\gamma \lesssim 0.05/t_a$. For all practically purposes, one can consider $\gamma = 0.05/t_a$ to be the condition for stability. A similar stability condition along with discussion for this argument is given in Ref. 11.

In order to quantify the amount of shear needed to stabilize $m=0$ modes, κ_{05} is defined to be the value of κ needed to reduce the growth rate to $\gamma = 0.05/t_a$. This quantity is shown in Fig. 7 for both Bennett and FuZE-like profiles. Bennett profiles can be *stabilized* to all ka for $\kappa \gtrsim 0.8$. This value is in agreement with that reported in Ref. 10. On the other hand, $\kappa > 4$ is required to *stabilize* sausage modes using FuZE-like profiles. This is a $5\times$ increase in shear compared to that needed to *stabilize* $m=0$ modes using Bennett profiles.

V. RESULTS FOR $m=1$ KINK MODES

The effects of shear on the $m=1$ kink mode have been analyzed previously in Ref. 9 and in Ref. 11. The growth rates of the two fastest growing $m=1$ modes considering Bennett profiles with $ka=10/3$, $R=3a$, and a quadratic axial velocity profile are given in Fig. 9 of Ref. 11. The two fastest $m=1$ modes obtained from our simulations using the same setup are shown in Fig. 8. With the exception of a difference in the normalization scales, the results shown in Fig. 8 here are in agreement with those in Fig. 9 of Ref. 11.

It is also stated in Ref. 11 that there are no significant differences for $m=1$ results between Bennett and Kadomtsev profiles. This again is in agreement with the results here. The effect of a linearly sheared axial velocity on the growth rate spectrum of the fastest growing $m=1$ kink modes for both Bennett and Kadomtsev profiles is illustrated in Fig. 9. The results are similar between these two sets of profiles for all values of κ considered.

Similar to how shear affects the growth rate spectrum for $m=0$ modes, increasing the shear pushes the maximum unstable $m=1$ mode to lower values of ka. However, the suppression of the growth rate with increasing κ is much less pronounced than that shown in Fig. 4 for $m=0$ sausage modes using Bennett profiles. While $\kappa \gtrsim 0.8$ is sufficient to *stabilize* all sausage modes using Bennett profiles, the growth time for $m=1$ modes is still comparable to the characteristic ion transit time t_a even for $\kappa = 3$. The maximum growth rate is peaked at $ka \approx 1$ with a value of $\gamma t_a \approx 0.7$ using both Bennett and Kadomtsev profiles with $\kappa = 3$.

The results presented in Figs. 8 and 9 are in agreement with the work by Arber *et al.* in Ref. 11. However, these results are in disagreement with the results presented by Shumlak and Hartman in Ref. 9, where the effect of a linearly sheared axial velocity on the $m=1$ kink instability using Kadomtsev profiles is studied. It is stated in Ref. 9 that complete stabilization of all $m=1$ modes occurs for $du_{z0}/dr = \kappa v_i/a \gtrsim 0.1kV_A^0$, where $V_A^0 \approx v_i$ is a characteristic Alfvén speed computed using the maximum value of magnetic field and maximum value of mass density ($V_A^0 = 0.96 v_i$ for Kadomtsev profiles). There is no mention of the values of k considered for the simulations in Ref. 9. In any case, the values of κ needed for stabilization based on the formula

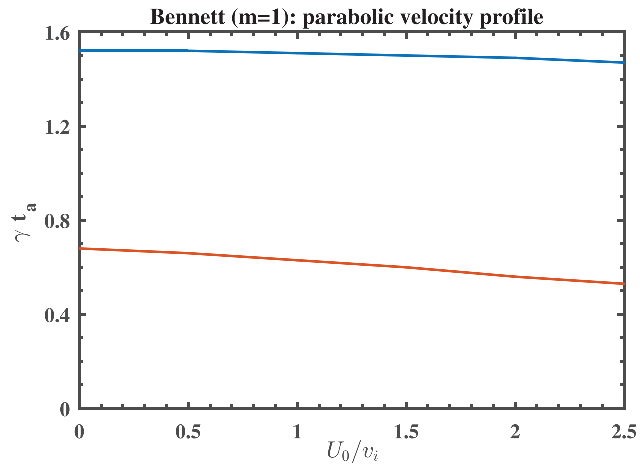

FIG. 8. Growth rates of the two fastest growing ($n=0$, 1) $m=1$ kink modes using Bennett profiles with $R=3a$, $ka=10/3$, and $u_{z0} = U_0(1-(r/R)^2)$. These results should be compared with Fig. 9 of Ref. 11.

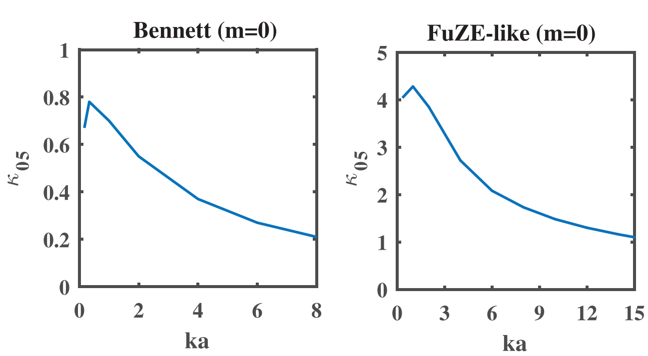

FIG. 7. Dimensionless shear strength κ needed to *suppress* $m=0$ growth rate to $0.05/t_a$ for Bennett (left) and FuZE-like (right) profiles with $u_{z0}/v_i = \kappa r/a$.

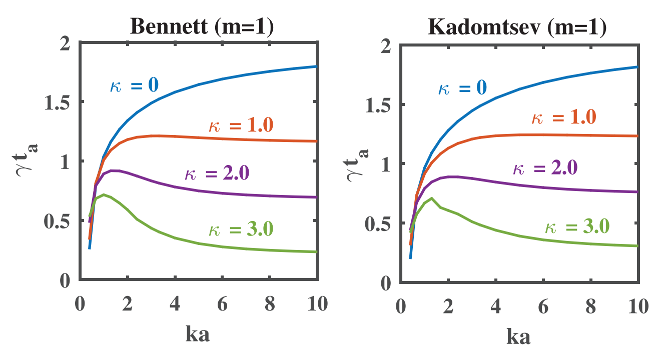

FIG. 9. Growth rate spectrum of fastest growing $m=1$ kink modes for different values of shear using Bennett (left) and Kadomtsev (right) profiles with $R=3a$ and $u_{z0}/v_i = \kappa r/a$.

from Ref. 9 is 1–2 orders of magnitude off for ka order unity and has an opposite scaling with ka in comparison to the results here.

A more detailed comparison of our results with those in Ref. 9 is made by looking at profiles of the radial Eulerian displacement element magnitude, $|\xi_r^E \equiv \tilde{u}_r/(-i\omega)|$. This quantity for $m=1$ using $R=3a$ and $ka=1$ with Kadomtsev profiles is shown in Fig. 10 for different values of κ. The value of $ka=1$ is chosen as it seems to agree with the same quantity shown in Fig. 3 of Ref. 9 for $\kappa=0$. However, the effect of shear on the radial displacement element reported here is much different than that from Ref. 9. That work found $|\xi_r^E|$ to be localized near the axis for a modest shear value of $\kappa=0.036$, which is declared to be the marginally stable κ corresponding to a zero growth rate. The results here show that indeed the profile gets squeezed toward the axis, but even with a shear value about 100 times larger it is not nearly as compressed on axis as is shown in Fig. 3 of Ref. 9. Moreover, the growth rate is only modestly affected going from $\gamma t_a \approx 1$ for no shear to $\gamma t_a \approx 0.7$ with $\kappa=3$.

The model in Ref. 9 has been criticized[17] for using an inconsistent boundary condition at $r=0$ for ξ_r^E. A zero gradient boundary condition is used at $r=0$ in Ref. 9 for $m=1$. However, the velocity profile in Ref. 9 is odd with respect to zero and, as discussed in Sec. III A and illustrated in Fig. 10, $\xi_r^E \sim \tilde{u}_r$ has a finite gradient at $r=0$ for this case. It is conjectured in Ref. 17 that this issue with the boundary condition at $r=0$ may be the reason for the disagreement of the results from Ref. 9 with those from Ref. 11. We cannot argue that this is indeed the case, but our analysis, as outlined above, supports the arguments expressed in Ref. 17 that the $r=0$ boundary condition in Ref. 9 appears to be inconsistent.

It is shown in Sec. IV that the characteristic growth rates for $m=0$ modes using FuZE-like profiles are much larger than that for Bennett profiles. On the other hand, the growth rates obtained using FuZE-like profiles with $m=1$ are approximately the same as those for Bennett and Kadomtsev profiles. The growth rate spectrum for FuZE-like profiles for $m=1$ using different values of κ are shown in Fig. 11.

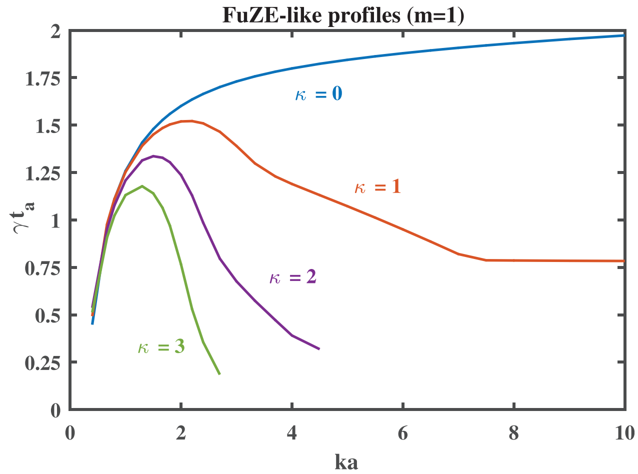

FIG. 11. Growth rate spectrum of fastest growing $m=1$ kink modes using FuZE-like equilibrium profiles with $R=3a$ and $u_{z0}/v_i = \kappa r/a$.

One difference for the FuZE-like profiles with $m=1$ compared to the other profiles is that it is seen to be more affected by shear at high ka. The growth rates quickly decay for $ka \gtrsim 3$ at $\kappa \gtrsim 2$ using FuZE-like profiles relative to the behavior for Bennett and Kadomtsev profiles for the same range of ka and κ. However, the growth rates for $m=1$ and $ka \approx 1$ are relatively unaffected by shear for all three sets of equilibrium profiles even for $\kappa=3$.

Another difference between the $m=1$ kink modes for FuZE-like profiles and the Kadomtsev/Bennett profiles is that the eigenmode is centered closer to $r=a$ for the FuZE-like profiles while it tends to be closer to the axis for the others. Contour plots of the pressure perturbation for $m=1$ using Kadomtsev and FuZE-like profiles are shown in Fig. 12 to illustrate this. The effect of a strong shear is to squeeze the eigenmode closer to the axis for Kadomtsev profiles while it remains localized in the vicinity of $r=a$ for FuZE-like profiles.

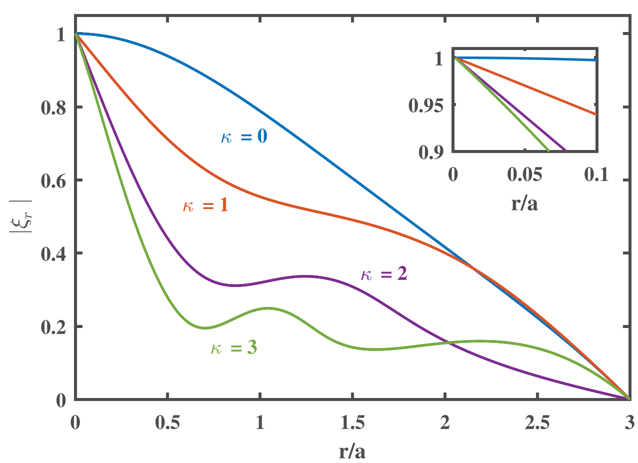

FIG. 10. Radial profiles of the radial Eulerian displacement element magnitude for $m=1$ kink mode using Kadomtsev equilibrium profiles with $ka=1$ and $u_{z0}/v_i = \kappa r/a$. These results are to be compared with Fig. 3 of Ref. 9. The inset figure is zoomed in near the axis showing the finite gradient of the solution at $r=0$ for $\kappa > 0$.

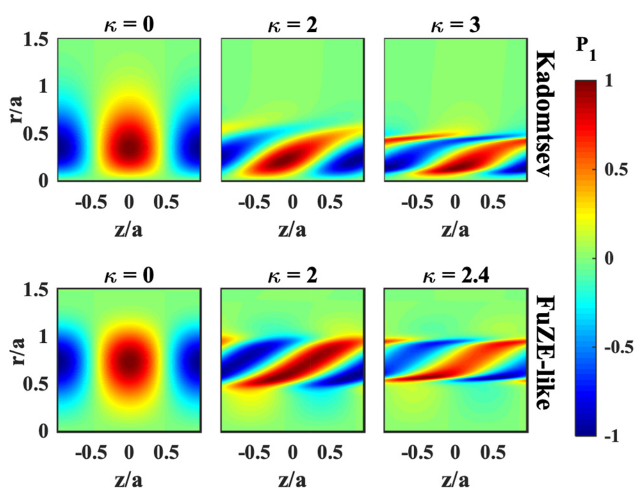

FIG. 12. Effect of non-dimensional shear strength κ on pressure perturbation for $m=1$ kink mode using Kadomtsev (top) and FuZE-like (bottom) profiles with $ka=10/3$ and $u_{z0}/v_i = \kappa r/a$.

VI. ANALYSIS OF RESULTS

The results presented above from solutions of the global eigenvalue problem for $m=0$ sausage and $m=1$ kink modes are analyzed here by (1) drawing connections with recent analytic studies of how a sheared velocity affects Rayleigh–Taylor (RT) type instabilities in neutral fluids and confined plasmas,[20,23] (2) examining the dispersion relation in the local limit ($ka \gg 1$) for the shear-free case, (3) understanding the behavior of the eigenmodes for large values of shear by examining the governing equations for $\kappa \gg 1$, and (4) presenting basic heuristic arguments that support the main results.

A. Effect of shear on $m=0$ modes

The effects of shear on $m=0$ sausage modes presented in Sec. IV are similar to how shear affects electrostatic interchange modes (IM) in magnetized plasmas and RT modes in neutral fluids.[20,23] The equations governing these RT type instabilities can be obtained as limiting forms of the ideal MHD equations that governs $m=0$ sausage modes. The drive for each of these instabilities is a pressure gradient that opposes some effective gravity force.[24]

Considering a particular set of equilibrium density profiles with a characteristic width a, analytic expressions for corrections to the shear-free growth rate and eigenmode for IM and RT modes are obtained in Refs. 20 and 23 by an appropriate small-parameter expansion of the eigenmode equation. It is found that there is a direct relationship between how much the eigenmode deforms and how much the growth rate is reduced resulting from a sheared axial flow—consistent with the results here discussed in Sec. IV. It is also shown analytically in Ref. 20 that for $ka \gg 1$ these modes are *suppressed* when $du_z/dr \gtrsim 2\sqrt{2}\gamma_0/(ka)$, where $\gamma_0 = \gamma_0(ka)$ is the shear-free growth rate spectrum. A different condition is obtained for $ka \to 0$.

A similar scaling with both γ_0 and ka for shear to strongly *suppress* $m=0$ sausage modes is found here using both Bennett and FuZE-like profiles. κ_{05} is defined in Sec. IV to be the value of du_{z0}/dr such that the growth rate falls to $0.05/t_a$. The ratio of κ_{05} to the shear-free growth rate is shown in Fig. 13. It is found for both sets of equilibrium profiles that this ratio can be expressed as $\kappa_{05}/\gamma_0 = C/(ka)^{\alpha}$, where C is an order unity constant and $\alpha \approx 1$. Specifically, $C=1.4$ and $\alpha = 0.80$ for Bennett profiles and $C=3.8$ and $\alpha = 0.86$ for FuZE-like profiles. This scaling holds for $1/3 \lesssim ka \lesssim 10$, which covers the practical range of ka values to consider.

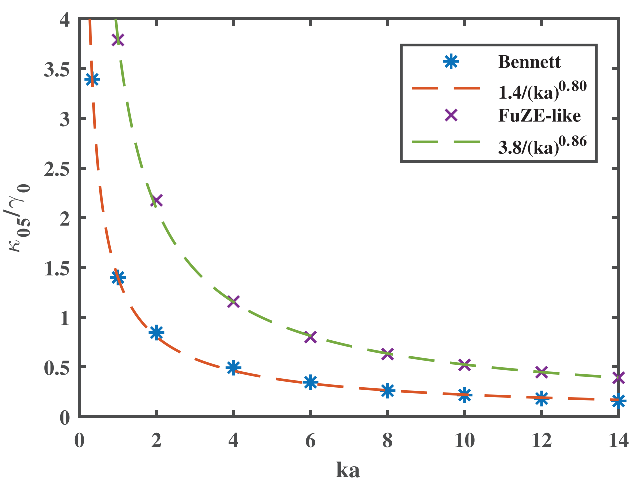

FIG. 13. Ratio of κ_{05} presented in Fig. 7 to shear-free $m=0$ growth rate γ_0 for Bennett and FuZE-like equilibrium profiles. These curves are fit to an equation of the form $C/(ka)^{\alpha}$.

B. Shear-free local analysis

The shear-free growth rate, and consequently the amount of shear needed to *stabilize* $m=0$ modes, can be sensitive to the equilibrium profiles. On the other hand, results for $m=1$ are less sensitive to the equilibrium profiles. This contrasting behavior can be understood by looking at the shear-free dispersion relation in the local limit where $k \equiv k_z \gg k_x \gtrsim |d\ln f_0/dr| \sim 1/a$ and f_0 is any equilibrium quantity. In this limit, the dispersion relation to leading order in $ka \gg 1$ obtained from Eqs. (23) and (24) is $C_3 = 0$. (C_3 is defined in Appendix B.) This results in a quadratic equation for $x \equiv \rho_0\omega^2$

$$a_0x^2 + b_0x + c_0 = 0. \tag{28}$$

The coefficients are

$$a_0 = \rho_0\left(v_A^2 + v_s^2\right), \tag{29}$$

$$b_0 = a_0\left(g + \frac{4\rho_0 v_A^2}{r^2} - F^2\right) - \rho_0 v_s^2\left(\frac{4\rho_0 v_A^2}{r^2} + F^2\right), \tag{30}$$

$$c_0 = -\rho_0 v_s^2 F^2\left(g + \frac{4\rho_0 v_A^2}{r^2} - F^2\right). \tag{31}$$

Definitions for each term in the above expressions for a_0, b_0, and c_0 can be found in Appendix B. It can be shown that $b_0^2 - 4a_0c_0$ is positive definite and thus the solutions of Eq. (28) for ω are either real or imaginary, but not mixed complex—consistent with properties of the global operator in the absence of a sheared flow.

The solution of Eq. (28) for $m=0$ reduces to[15,16]

$$\omega^2 = \frac{4v_i^2}{r^2}\left(\frac{d\ln P_0}{d\ln r} + \frac{4\Gamma}{2+\Gamma\beta_0}\right), \quad \text{for } m=0. \tag{32}$$

The condition in parenthesis is precisely the Kadomtsev criteria for sausage mode stability given in Eq. (6). For $m>0$, the solution of Eq. (28) is

$$\omega^2 = \frac{-b_0 \pm \sqrt{b_0^2 - 4a_0c_0}}{2a_0\rho_0}, \quad \text{for } m>0. \tag{33}$$

The condition for Eq. (33) to be positive and thus stable is $g + 4\rho_0 v_A^2/r^2 - F^2 \le 0$, which is precisely the Kadomtsev stability condition for $m>0$ given in Eq. (7).

The positive imaginary parts of the solutions of Eqs. (32) and (33) are shown in Fig. 14 for each set of equilibrium profiles considered in this work. The first important thing to note is that the maximum value from the local dispersion agree well with the growth rates from the global solutions for large ka for each set of equilibrium profiles and both $m=0$ and $m=1$. In particular, the maximum values for $m=0$ are $\gamma t_a \approx 0.83$ using Bennett profiles and $\gamma t_a \approx 3.3$ for FuZE-like profiles. The maximum value for $m=1$ is $\gamma t_a \approx 2$ for all three sets of equilibrium profiles.

The local solution turns out to contain more useful information. In particular, it also shows the radial location where the drive for the

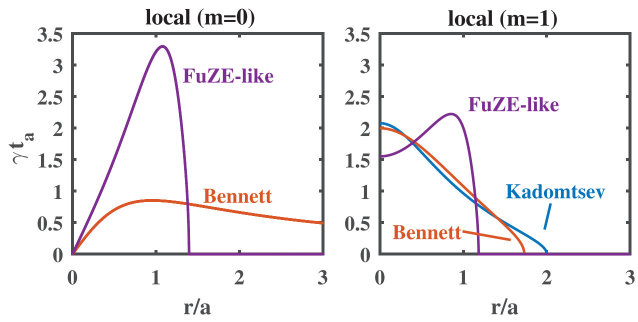

FIG. 14. Radial profiles of positive imaginary parts (growth rate) of the local dispersion relation for Kadomtsev (blue), Bennett (red), and FuZE-like (purple) equilibrium profiles. Results for $m = 0$ from Eq. (32) are shown on the left and Results for $m = 1$ from Eq. (33) are shown on the right.

instability is peaked. For $m = 0$, this location is $r \approx a$ for both Bennett and FuZE-like profiles. The drive for Bennett and Kadomtsev profiles is both peaked on axis for $m = 1$. However, the drive for $m = 1$ using FuZE-like profiles is peaked around $r = a$, same as it is for $m = 0$. The radial locations where the eigenmodes from the global solutions are peaked, as seen in Figs. 6 and 12, are in agreement with this analysis.

The local description here for the shear-free case uses the classic WKB approximation with $1/ka$ as the small parameter. It is tempting to do the same analysis with shear retained, in which case one would arrive at the same results in Eqs. (32) and (33) with ω replaced by $\omega' = \omega - ku_{z0}$. However, the WKB approximation is not valid with a sheared axial flow because of how the shear affects the shape of the eigenfunction.[19] A different small parameter has to be considered for an appropriate analysis of the linear equations that agrees with the certain asymptotic limits of the global solution when a finite shear is included.[19,20] Furthermore, the linear system loses its Hermitian properties when a sheared axial flow is included and the eigenmodes are affected in such a way that the existence of non-modal solutions is possible.[25,26] An asymptotic analysis of the governing equations with finite shear and analysis of non-modal solutions are both beyond the scope of the present paper.

C. Different behavior of eigenmodes for $m = 0$ and $m = 1$ with large shear

The eigenmodes for both $m = 0$ and $m = 1$ get stretched axially and squeezed radially for large values of shear, as seen in Figs. 6 and 12. However, one distinct feature of the behavior of the $m = 1$ eigenmode for large shear is that the stretching seems to occur in two localized regions in r while the eigenmodes for $m = 0$ become localized at a single location in r. These locations correspond to where the coefficients preceding the radial derivative terms in Eqs. (23) and (24) go to zero.[11] This coefficient is AS for both equations. A and S are defined in Appendix B.

The imaginary part of $\omega' = \omega - ku_{z0}$ goes to zero for infinite shear and the real part converges to $k\kappa v_i r_0/a$, where r_0 is a radial location near where the eigenmode is centered. A and S can both independently be zero and, in general, there are four distinct roots given by

$$\omega'^2 = \frac{F^2}{\rho_0}, \quad \text{and} \quad \omega'^2 = \frac{v_s^2}{v_A^2 + v_s^2}\frac{F^2}{\rho_0}. \tag{34}$$

Both of the conditions in Eq. (34) reduce to the same thing near the axis since $v_A^2 \to 0$ as $r \to 0$. Furthermore, the solutions are trivial for sausage modes since $F = 0$ for $m = 0$, in which case the real part of ω'^2 is zero at $r = r_0$.

For the equilibrium profiles considered in this work, it is found that both conditions in Eq. (34) give approximately the same values. The latter expression in Eq. (34) is compared with the real part of ω'^2 in Fig. 15 for Kadomtsev profiles and FuZE-like profiles for $m = 1$. The real part of ω'^2 shown in this figure corresponds the largest κ values shown in Fig. 12, which is $\kappa = 3.0$ for the Kadomtsev profiles and $\kappa = 2.4$ for the FuZE-like profiles. The radial locations where these curves cross are seen to correspond to the inner and outer radii of the perturbed pressure contours shown in Fig. 12 for both sets of profiles.

D. Heuristic arguments

The only modification to the governing equations, Eqs. (23) and (24), when a sheared axial flow is included is $\omega \to \omega' = \omega - ku_{z0}(r)$ in the coefficients. Based on this, it can be argued that the effects of a sheared flow start to become important when the change in ω' across some characteristic radial width of the eigenmode, δr, becomes comparable to $\omega = i\gamma_0$ from the shear-free solution. In other words, the condition is

$$\delta r\left|\frac{d\omega'}{dr}\right| \gtrsim \gamma_0 \quad \Rightarrow \quad \left|\frac{du_{z0}}{dr}\right| \gtrsim \frac{\gamma_0}{k\delta r}. \tag{35}$$

While δr depends on the equilibrium profiles as well as ka, typical values are such that $\delta r \lesssim a$. The condition given in Eq. (35) is in good agreement with the results presented in this paper in terms of the magnitude of shear needed to affect the system as well as the scaling with the shear-free growth rate γ_0 and axial wavenumber k.

The relation in Eq. (35) physically corresponds to the shear causing fluid elements displaced radially by δr to become decorrelated by a length scale of k^{-1} over a time shorter than or comparable to the shear-free growth time.[12] The correlation time is defined as $\tau_c = 1/(k\delta r|du_{z0}/dr|)$. The dimensionless ratio $(\gamma_0\tau_c)^2$ is known in the tokamak and fluid communities as the Richardson number when $k\delta r = 1$. It is important to mention that the heuristic arguments presented here are just for the shear to have an effect. It does not say what the effect is, which depends on the system and

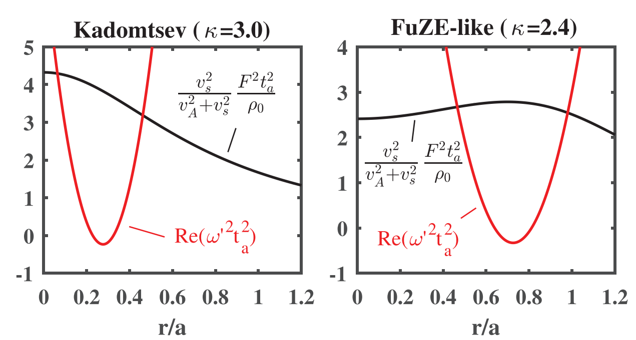

FIG. 15. Radial profiles of left and right sides of the latter expression in Eq. (34) for Kadomtsev (left) and FuZE-like (right) profiles are shown in black. The red curves are obtained using $\kappa = 3.0$ for the Kadomtsev profiles and $\kappa = 2.4$ for the FuZE-like profiles (see Fig. 12).

instability of interest.[12] Furthermore, for certain instabilities, the Richardson number alone is not always a sufficient condition to describe when shear is important.[20]

VII. CONCLUSIONS

Results from a global eigenmode analysis of the sheared-flow Z-pinch are presented in this paper. The results are obtained via a direct eigen-decomposition of the operator corresponding to the linear ideal MHD equations. Both $m=0$ sausage and $m=1$ kink modes are studied and several different equilibrium profiles are examined. This includes the well-known Kadomtsev and Bennett profiles as well as a new class of equilibrium solutions that can be fit to recently reported measurements from the FuZE device.[7] The effects of a linearly sheared axial velocity profile are studied. Although not presented, different profiles for the axial velocity, such as parabolic and sin^2, have also been considered. The results are qualitatively the same as that for a linear profile.

It is shown that the amount of shear required to *stabilize* $m=0$ modes can be expressed as $du_{z0}/dr \gtrsim C\gamma_0/(ka)^{\alpha}$, where $\gamma_0 = \gamma_0(ka)$ is the profile-specific growth rate spectrum with no shear, C is an order unity constant, and $\alpha \approx 1$. It is shown that four to five times more shear is needed to *stabilize* $m=0$ modes when using FuZE-like profiles compared to that needed for Bennett profiles, consistent with a four to five times increase in the shear-free growth rates. The shear-free growth rates scale with characteristic values of $|d\ln P/d\ln r + 4\Gamma/(2+\Gamma\beta)|$, which is relatively small for Bennett profiles. Consequently, the amount of shear needed to *stabilize* $m=0$ modes is relatively small for Bennett profiles.

The effects of shear on $m=1$ kink modes is found to be less sensitive to the choice of equilibrium profiles compared to $m=0$, as are the shear-free growth rates. However, even for $du_{z0}/dr = 3v_i/a$, the growth rates for $m=1$ and $ka \approx 1$ are not strongly suppressed for all profiles considered in this work. This result is in contrast to the commonly accepted theory that says complete stabilization of $m=1$ modes is achieved for $du_{z0}/dr \gtrsim 0.1kv_i$ (see Ref. 9). Not only do our results suggest that the shear needed to *suppress* $m=1$ modes is 1–2 orders of magnitude larger than that from Ref. 9, but the scaling with k is completely different. It is shown here that higher-ka modes are more strongly affected by shear for $m=1$ same as they are for $m=0$. The behavior of the global solutions presented here is in line with expectations based on examination of the dispersion relation in the local limit ($ka \gg 1$) in the absence of shear, analysis of the governing equations in the limit of large shear, and basic heuristic arguments.

While the results here for $m=1$ do not align with those in Ref. 9, they are in agreement with those in Ref. 11. Our main conclusion here is the same as that in Ref. 11—if flows with $du_{z0}/dr \lesssim v_i/a$, such as those measured in experiments, are responsible for the long stabilization times for some Z-pinch systems, then it should be a nonlinear process[10,11] or related to non-ideal/kinetic physics, such as finite-Larmor radius effects.[18,27,28] While flows with $du_{z0}/dr \lesssim v_i/a$ are not sufficient to provide complete linear stability in the ideal limit, such flows could, however, potentially alter the nonlinear saturation of the instability such that the system is nonlinearly stable.[10] In particular, the radial location of the eigenmode for the FuZE-like profiles considered here depends on the pressure floor used in the profiles since this affects the logarithmic pressure gradient. Lowering the pressure floor will cause the growth rate to increase, thus increasing the shear needed for linear stabilization, but it will also push the mode further out on the periphery of the pinch where it may saturate at a low enough level that it does not disrupt the macroscopic stability of the pinch. Understanding and quantifying effects of this nature and the interplay with a sheared flow require nonlinear studies.[29]

ACKNOWLEDGMENTS

This work was performed under the auspices of the U.S. Department of Energy by Lawrence Livermore National Laboratory under Contract No. DE-AC52–07NA27344 and supported by the Laboratory Directed Research and Development Program (No. 18-ERD-007). No. LLNL-JRNL-814236.

APPENDIX A: FUZE-LIKE PROFILES

Density and temperature profiles from FuZE[7] can be fit using expressions of the form[29]

$$n = n_0 + n_1 \exp\left(-c_0x^2 - c_1x^4\right), \tag{A1}$$

$$T = T_0 + T_1 \exp\left(-c_2x^2\right), \tag{A2}$$

where $x = r/a$ and $a = 0.37$ cm. The coefficients are $c_0 = 0.4278$, $c_1 = 1.059$, $c_2 = 1.097$, $n_0 = 0.05 \times 10^{17}/\mathrm{cm}^3$, $n_1 = 0.855 \times 10^{17}/\mathrm{cm}^3$, $T_0 = 0.4015$ keV, and $T_1 = 1.089$ keV. Using Eqs. (A1) and (A2), the pressure $P = 2nT$ is of the form given in Eq. (9). The general solution of Eq. (5) using the pressure profile in Eq. (9) is

$$\frac{(xB_\theta)^2}{2\mu_0} = -x^2(P - P_{00}) + \sum_{n=1}^{3} P_n\psi_n(x), \tag{A3}$$

where the function $\psi_n(x)$ for $b_n > 0$ is

$$\psi_n(x) = \frac{\exp\left(\dfrac{a_n^2}{4b_n}\right)}{2\sqrt{b_n/\pi}}\left(\mathrm{Erf}\left[\frac{a_n + 2b_nx^2}{2\sqrt{b_n}}\right] - \mathrm{Erf}\left[\frac{a_n}{2\sqrt{b_n}}\right]\right). \tag{A4}$$

If $b_n = 0$, then $\psi_n(x)$ becomes

$$\psi_n(x) = \frac{1}{a_n}\left(1 - \exp\left[-a_nx^2\right]\right). \tag{A5}$$

The pressure profile here is obtained using a spatially varying temperature. However, T is assumed constant for the simulations such that the density profile is the same as the pressure profile, consistent with the other sets of profiles considered. This change has a negligible effect on the results because (1) the temperature variation is small and (2) stability using ideal MHD only depends on P [see Eqs. (6) and (7)].

APPENDIX B: COEFFICIENTS IN EQS. (23) AND (24)

Using $v_A^2 = B_{00}^2/(\rho_0\mu_0)$ and $v_s^2 = \Gamma P_0/\rho_0$, the coefficients in Eqs. (23) and (24) are

$$A = \rho_0\omega'^2 - F^2, \quad F^2 = \frac{m^2}{r^2}\rho_0v_A^2, \tag{B1}$$

$$S = \rho_0^2\omega'^2v_A^2 + \rho_0v_s^2A, \tag{B2}$$

$$rC_1 = 2\rho_0 v_A^2 A\left(\rho_0\omega'^2 - \frac{m^2}{r^2}\rho_0 v_s^2\right), \tag{B3}$$

$$C_2 = \rho_0^2\omega'^4 - \left(\frac{m^2}{r^2} + k^2\right)S, \tag{B4}$$

$$C_3 = AS(A+g) + 2\frac{\rho_0 v_A^2}{r^2} rC_1, \tag{B5}$$

$$g = r\frac{d}{dr}\left(\frac{\rho_0 v_A^2}{r^2}\right) = -\frac{2}{r}\left(\frac{dP_0}{dr} + \frac{2\rho_0 v_A^2}{r}\right), \tag{B6}$$

where $\omega' = \omega - ku_{z0}$. The latter expression for g results from making use of radial force balance from Eq. (5).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

[1] U. Shumlak, J. Appl. Phys. **127**, 200901 (2020).
[2] U. Shumlak, B. A. Nelson, E. L. Claveau, E. G. Forbes, R. P. Golingo, M. C. Hughes, R. J. Oberto, M. P. Ross, and T. R. Weber, Phys. Plasmas **24**, 055702 (2017).
[3] P. Reynolds and J. Craggs, London Edinburgh Dublin Philos. Mag. J. Sci. **43**, 258 (1952).
[4] I. Kurchatov, J. Nucl. Energy (1954) **4**, 193 (1957).
[5] U. Shumlak, R. P. Golingo, B. A. Nelson, and D. J. Den Hartog, Phys. Rev. Lett. **87**, 205005 (2001).
[6] U. Shumlak, B. A. Nelson, R. P. Golingo, S. L. Jackson, E. A. Crawford, and D. J. D. Hartog, Phys. Plasmas **10**, 1683 (2003).
[7] Y. Zhang, U. Shumlak, B. A. Nelson, R. P. Golingo, T. R. Weber, A. D. Stepanov, E. L. Claveau, E. G. Forbes, Z. T. Draper, J. M. Mitrani, H. S. McLean, K. K. Tummel, D. P. Higginson, and C. M. Cooper, Phys. Rev. Lett. **122**, 135001 (2019).
[8] R. P. Golingo, U. Shumlak, and B. A. Nelson, Phys. Plasmas **12**, 062505 (2005).
[9] U. Shumlak and C. W. Hartman, Phys. Rev. Lett. **75**, 3285 (1995).
[10] I. Paraschiv, B. Bauer, I. Lindemuth, and V. Makhin, Phys. Plasmas **17**, 072107 (2010).
[11] T. D. Arber and D. F. Howell, Phys. Plasmas **3**, 554 (1996).
[12] P. W. Terry, Rev. Mod. Phys. **72**, 109 (2000).
[13] M. G. Haines, Plasma Phys. Controlled Fusion **53**, 093001 (2011).
[14] K. Tummel, D. P. Higginson, A. J. Link, A. E. W. Schmidt, D. T. Offermann, D. R. Welch, R. E. Clark, U. Shumlak, B. A. Nelson, R. P. Golingo, and H. S. McLean, Phys. Plasmas **26**, 062506 (2019).
[15] B. Kadomtsev, Sov. Phys. JETP-USSR **10**, 780 (1960).
[16] J. R. Angus, M. Dorf, and V. I. Geyko, Phys. Plasmas **26**, 072505 (2019).
[17] T. D. Arber and D. F. Howell, Phys. Rev. Lett. **76**, 2198 (1996).
[18] V. I. Geyko, M. Dorf, and J. R. Angus, Phys. Plasmas **26**, 062114 (2019).
[19] E. S. Benilov, V. Naulin, and J. J. Rasmussen, Phys. Fluids **14**, 1674 (2002).
[20] Y. Zhang, S. I. Krasheninnikov, and A. I. Smolyakov, Phys. Plasmas **27**, 020701 (2020).
[21] P. Guzdar, P. Satyanarayana, J. Huba, and S. Ossakow, Geophys. Res. Lett. **9**, 547, https://doi.org/10.1029/GL009i005p00547 (1982).
[22] S. Appl and M. Camenzind, Astron. Astrophys. **256**, 354 (1992), see https://ui.adsabs.harvard.edu/abs/1992A&A...256..354A.
[23] Y. Zhang, S. I. Krasheninnikov, and A. I. Smolyakov, Contrib. Plasma Phys. **60**, e201900098 (2020).
[24] J. R. Angus and S. I. Krasheninnikov, Phys. Plasmas **21**, 112504 (2014).
[25] T. Tatsuno, F. Volponi, and Z. Yoshida, Phys. Plasmas **8**, 399 (2001).
[26] V. S. Mikhailenko, E. E. Scime, and V. V. Mikhailenko, Phys. Rev. E **71**, 026306 (2005).
[27] T. D. Arber, M. Coppins, and J. Scheffel, Phys. Rev. Lett. **72**, 2399 (1994).
[28] T. D. Arber, P. G. F. Russell, M. Coppins, and J. Scheffel, Phys. Rev. Lett. **74**, 2698 (1995).
[29] V. I. Geyko, J. R. Angus, and M. A. Dorf, "Simulation of fuze axisymmetric stability using gyrokinetic and extended-mhd models," arXiv:2011.02767 (2020).